	ITER, SIKSHA 'O' ANUSANDHAN **Deemed to be University**		**LESSON PLAN**
Programme	**B. Tech.**	Academic Year	**2025-26**
Department	**CSE**	Semester	**8th**
Coordinator	**Dr. Gayatri Mohapatra**	Grading Pattern	**6**
Subject Code	**OPL4001**		
Subject Name	**RENEWABLE ENERGY PRINCIPLES**		

Text Books(s):

(1) Text book-Renewable Energy, Power for a sustainable future, By Stephan Peake, OUP

Course Format: 3 Class/week, 1 hr/Class

Course Outcomes	Students will be able to	
	CO1	Understand different sources of energy and their effects in developing aware ness on impacts of global warming.
	CO2	Solar Thermal energy and Understanding how Solar radiation reaches the earth's surface.
	CO3	Solar Photo Voltaic system, its type and its integration.
	CO4	Hydro energy, types of hydroelectric plant its environmental impact.
	CO5	Tidal power, Power generation from tidal barrages and its environmental impact.
	CO6	Wind energy, different types of wind turbine. Power and energy from wind turbine and its environmental impact.

Lecture/	**Lessons/Topics to be covered**	**Book Reference (sections)**	**Mapping with COs**	**Assignment and Quizzes**
Week #1:				
Lecture#1	Introduction to renewable energy sources.	1.1	**CO1**	
Lecture#2	World energy supplies. Categorization of energy in different sectors.	1.2	**CO1**	
Lecture#3	Fossil fuels. Adverse effects of fossil fuels on the environment.	1.3	**CO1**	

Week #2:				
Lecture#4	Greenhouse gases and global warming. Its effect on the environment.	1.3	**CO1**	
Lecture#5	Different types of Renewable energy sources	1.4	**CO1**	
Lecture#6	Solar energy	4.1,4.2	**CO2**	
Week #3:				
Lecture#7	Solar radiation reaches the earth's surface.	4.3	**CO2**	
Lecture#8	Active Solar heating with examples.	4.6	**CO1**	
Lecture#9	Passive solar heating with examples.	4.7	**CO2**	**Assignment 1**
Week #4:				
Lecture#10	Numerical on solar radiation geometry.		**CO2**	
Lecture#11	Solar thermal engine and Electricity generation	4.9	**CO2**	
Lecture#12	Thermal applications, Solar pond, Solar Cooker.	4.10	**CO2**	
Week #5:				
Lecture#13	Solar thermal applications. Rooftop water solar heater	4.10	**CO2**	
Lecture#14	Solar radiation measuring instruments Pyranometer, Pyro heliometer and Sunshine Recorder	4.12	**CO2**	
Lecture#15	Explanation about different types of Solar Collectors	4.12	**CO2**	
Week #6:				
Lecture#16	History of PV cell.	5.2	**CO3**	**Quiz 1**
Lecture#17	Construction of different types of available Silicon solar cells	5.2	**CO3**	
Lecture#18	Types of PV cells	5.3	**CO3**	
Week #7:				
Lecture#19	Principle of generation of electricity	5.3	**CO3**	
Lecture#20	Electrical characteristics of Photo Voltaic cells.	5.7	**CO3**	**Assignment 2**
Lecture#21	Power and efficiency calculation of solar pv	5.7	**CO3**	
Week #8:				
Lecture#22	Standalone PV system	5.8	**CO3**	
Lecture#23	Grid connected PV	5.9	**CO3**	

Lecture#24	Cost of energy from PV, its integration and impact on the environment.	5.10, 5.12	**CO3**	
Week #9:				
Lecture#25	Energy economics of pv systems.	5.12	**CO3**	**Quiz 2**
Lecture#26	Hydro energy	7.3	**CO4**	
Lecture#27	Types of Hydroelectric plant.	7.7	**CO4**	
Week #10:				
Lecture#28	Generation of electricity by hydropower	7.9	**CO4**	
Lecture#29	Technical Factors behind Hydro Power generation	7.10	**CO4**	
Lecture#30	Tidal Power	8.1	**CO5**	**Assignment 3**
Week #11:				
Lecture#31	Tidal Barrage, Potential and project	8.6	**CO5**	
Lecture#32	History of wind energy. Classification of wind turbines	10.1	**CO6**	
Lecture#33	Different components of wind turbine.	10.3	**CO6**	**Quiz 3**
Week #12:				
Lecture#34	Power Vs wind speed characteristics	10.5	**CO6**	
Lecture#35	Future investment in Renewable energy	11.5	**CO6**	
Lecture#36	Revision			